Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equation for buck SMPS in continuous mode $\frac{V_O}{V_I} = \delta$

Voltage ratio equations for boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{1}{1-\delta}$

Voltage ratio equations for buck-boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{-\delta}{1-\delta}$

Three-Phase Systems (Balanced)

Line/Phase Voltages and Currents

Star $V_L = \sqrt{3}\, V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\, I_P$

Power $P_{3\phi} = 3\, V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Transformers

Ideal transformer $\bar{V}_2 = k_N \bar{V}_1$; $\bar{I}'_2 = k_N \bar{I}_2$

Impedances referral $\bar{Z}'_2 = {\bar{Z}_2}/{k_N{}^2}$

Voltage regulation $\Delta V_2 = \frac{\left|V_2^{o/c}\right| - |V_2^{load}|}{\left|V_2^{o/c}\right|}$

Induction Machines

$\omega_S = {\omega_E}/{P}$; $\quad s = \frac{\omega_S - \omega_R}{\omega_S}$; $\quad T_{EM} = \frac{3\, I'^2_R R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $\quad L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $\quad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $\quad G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $\quad L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $\quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $\quad G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}}\cos(\theta) - \frac{V_R V_S}{Z_{SR}}\cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}}\sin(\theta) - \frac{V_R V_S}{Z_{SR}}\sin(\theta+\delta)$$

Power Flow assuming inductance-dominated impedance

$$P_S \simeq \frac{V_R V_S}{X_{SR}}\sin(\delta) \qquad Q_S \simeq \frac{V_S}{X_{SR}}\left(V_S - V_R\cos(\delta)\right)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\, P_S + X\, Q_S}{|V_S|}$

Question 1 – Transformers. This question is worth 25 marks.

a) Some written questions on transformers

i) Explain whether a transformer can operate with a direct current (DC) supply. In your answer, provide a clear justification for your reasoning. [2]

Cannot operate on DC. [1]

No changing magnetic flux, so no induced emf. [1]

ii) Name a transformer that allows adjustment of voltage levels and explain how this adjustment is carried out. [2]

Tap changing transformers. [1]
Voltage is controlled by selecting different taps/number of turns on the winding. [1]

b) A 50 Hz single-phase step-down transformer has a rated primary voltage of 2.4 kV. The rated primary current is 41.67 A and the rated secondary current is 416.7 A

i) Calculate the turns ratio. [1]

Turn ratio $k = \frac{N_\mathrm{p}}{N_\mathrm{s}} = \frac{I_\mathrm{s}}{I_\mathrm{p}} = 10.$ [1]

ii) Calculate the rated secondary voltage. [1]

Secondary voltage $V_\mathrm{s} = \frac{V_\mathrm{p}}{10} = \frac{2400}{10} = 240\ \mathrm{V}.$ [1]

iii) Hence determine the rated apparent power of the transformer. [1]

Rated apparent power $S = V_\mathrm{s} I_\mathrm{s} = 240 \times 416.7 \approx 1.0 \times 10^5 \mathrm{VA} = 100\mathrm{kVA}.$ [1]

c) For the transformer in (b), rated voltage is applied to the primary side and the secondary side is open circuit. The open-circuit test results are:

Input voltage	$V_0 = 2.4\ \mathrm{kV}$
Input current	$I_0 = 0.5\ \ \mathrm{A}$
Input power	$P_0 = 724\mathrm{W}$

i) Calculate the **approximate** magnetizing loss of the transformer. [2]

In the open-circuit test, the secondary is open, so copper loss is very small.

$$P_m \approx P_0 = 724\mathrm{W}$$ [2]

ii) Calculate the power factor stating whether it is lagging or leading. [1]

Power factor

$$cos\phi_0 = \frac{P_0}{V_0 I_0} = \frac{724}{2400 \times 0.5} = 0.603$$ [1]

Lagging.

iii) Calculate the magnetizing resistance R_M. [2]

$$R_M = \frac{V_0^2}{P_0} = \frac{2400^2}{724} \approx 7.96\ \mathrm{k\Omega}$$ [2]

iv) Calculate the magnetizing reactance X_M. [2]

$$I_{\mathrm{w}} = \frac{P_0}{V_0} = \frac{724}{2400} \approx 0.302\mathrm{A}$$

$$I_{\mathrm{m}} = \sqrt{I_0^2 - I_w^2} = \sqrt{0.5^2 - 0.302^2} \approx 0.399\mathrm{A} \quad [1]$$

$$X_M = \frac{V_0}{I_{\mathrm{m}}} = \frac{2400}{0.399} = 6.015 \times 10^3\ \Omega$$

$$X_M \approx 6.015\mathrm{k}\Omega \quad [1]$$

d) For the transformer in (b), rated voltage is applied to the primary side. Under load, the secondary current lags the secondary terminal voltage by 11.3°, and the secondary current magnitude is 410.0A. The transformer input power is 93.9 kW, and the efficiency is 98.5%.

i) Calculate the output apparent power. [2]

$$P_{\mathrm{out}} = \eta P_{\mathrm{in}} = 0.985 \times 93.9 = 92.49\mathrm{kW}$$

$$S_{\mathrm{out}} = \frac{P_{\mathrm{out}}}{\cos 11.3^\circ} = \frac{92.49}{\cos 11.3^\circ} \approx 94.3\mathrm{kVA} \quad [2]$$

ii) Calculate the real and reactive power absorbed by the load. [3]

$$P_L = P_{out} = 92.49\mathrm{kW} \quad [1]$$

$$Q_L = P_L \times \tan 11.3^\circ \approx 18.5\mathrm{kVAr} \quad [2]$$

iii) Calculate the secondary terminal voltage. [3]

$$S_{\mathrm{out}} = V_s I_s$$

$$V_s = \frac{S_{\mathrm{out}}}{I_s} = \frac{94.3\times10^3}{410} \approx 230\mathrm{V} \quad [3]$$

iv) Determine the load impedance connected to the secondary side. [3]

$$Z_L = \frac{V_s}{I_s} = \frac{230}{410} \approx 0.561\Omega$$

The current lags by 11.3°, so

$$Z_L \approx 0.561\angle 11.3^\circ\ \Omega \quad [3]$$

Or

$$Z_L \approx 0.55 + j0.110\ \Omega$$

Question 2 – Three-Phase Synchronous Generators and Inverters. This question is worth 20 marks.

a) A three-phase, 600 kVA synchronous generator is star connected to an infinite bus at 3.3kV (line-to-line) and 50Hz. The generator has 2 pole pairs. (Assume balanced steady-state operation. Unless otherwise stated, neglect mechanical losses, iron losses and stray losses. Also neglect armature resistance.)

i) Calculate the synchronous speed of rotation in revolutions per minute. [2]

$$n_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

ii) At synchronous speed, the shaft torque is 3581 N · m. Determine the generator real power output. [2]

$$P = T\omega_s = \frac{2\pi n_s T}{60} = \frac{2\pi \times 1500 \times 3581}{60} = 562.50kW$$

iii) The shaft torque is then increased to 3760 N · m while the machine remains connected to the infinite bus. Determine the new real power output of the generator. [2]

Speed does not change since it is a synchronous generator.

$$P = T_{new}\omega_s = \frac{2\pi \times 1500 \times 3760}{60} = 590.62kW$$

iv) For this increased torque condition in (ii), the power factor is unity seen at the terminal and the generator internal emf is 2.1 kV RMS per phase. Calculate the synchronous reactance X_s per phase. [2]

$$V_s = \frac{V_L}{\sqrt{3}} = \frac{3300}{\sqrt{3}} = 1905.26 \text{ V}$$

$$I = \frac{P}{\sqrt{3}V_L} = \frac{590.62 \times 10^3}{\sqrt{3} \times 3300} = 103.33 \text{ A}$$

$\bar{E} = \bar{V}_s + jX_s\bar{I}$, power factor is unity, therefore $E^2 = {V_s}^2 + (X_s I)^2$

$$X_s = \frac{\sqrt{E^2 - {V_s}^2}}{I} = X_s = \frac{\sqrt{2100^2 - 1905.26^2}}{103.33} = 8.55\Omega/\text{phase}$$

v) The excitation current is increased by 10%. Assume that the internal EMF is directly proportional to the excitation current in this operating region. If the internal EMF before the change is 2.1 kV RMS per phase, and the real power output remains unchanged, determine the magnitude of the reactive current component per phase, and state whether it is supplied or absorbed by the generator seen at the terminal. [3]

$$\sin\delta = \frac{PX_s}{3V_s E'} = \frac{590.62 \times 10^3 \times 8.55}{3 \times 1905.26 \times 2310} \approx 0.382$$

$$\delta \approx 22.49°$$

$$\bar{I} = \frac{\overline{E'} - \bar{V}_s}{jX_s} = \frac{E'\sin\delta}{X_s} - j\frac{E'\cos\delta - V_s}{X_s}$$

$\bar{I} = I_P + jI_Q$, therefore the magnitude of I_Q is

$$|I_Q| = \left|-\frac{E'\cos\delta - V_s}{X_s}\right| = \left|-\frac{2310\cos 22.49° - 1905.26}{8.55}\right| = 26.79\text{A/phase}$$

Current lagging terminal voltage under the generator convention, therefore it is supplied by the generator.

b) A three-phase permanent magnet synchronous motor (PMSM) is driven by a three-phase inverter.

i) A sinusoidal PWM inverter operates with a switching frequency of 10 kHz and an output fundamental frequency of 50 Hz. Determine the modulation ratio. Is the obtained frequency modulation ratio reasonable? Briefly justify your answer. [2]

$$c = \frac{f_s}{f_1} = \frac{10000}{50} = 200$$

It is reasonable, since $f_s >> f_1$, the output waveform is closer to a sine wave after the low pass filter. Harmonics will concentrate near higher frequencies, making them easier to filter out. Ideally c should be at least 100.

ii) The inverter has dc input voltage (rail-to-rail) of 1200 V, which is equivalent to $V_{DC+}=V_{DC-}=600V$. Write the three sinusoidal PWM modulator functions required to generate a 50 Hz three-phase inverter voltage with a phase fundamental peak voltage of 300 V. [2]

Take phase a as an example,

$$v_a(t) = V_m \sin(2\pi f t + \varphi_a) = V_{DC}\, m_a(t)$$

$$m_a(t) = \frac{V_m}{V_{DC+}} \sin(2\pi f t + \varphi_a) = \frac{300}{600} \sin(100\pi t + 0) = 0.5\sin(100\pi t)$$

Similarly,

$$m_b(t) = 0.5\sin\left(100\pi t - \frac{2\pi}{3}\right)$$

$$m_c(t) = 0.5\sin\left(100\pi t + \frac{2\pi}{3}\right)$$

iii) At 1500 RPM, the motor back-EMF is 180 V line-to-line RMS. Assume the stator impedance voltage drop is negligible, so that the inverter fundamental output line-to-line RMS voltage is equal to the motor back-EMF. If the dc input voltage is 1200 V and the modulation index is 0.90, determine the motor speed at which this modulation index is required. [2]

$$V_{p_new} = m_a \times \frac{V_{DC}}{2} = 0.9 \times 600 = 540 \text{ V}$$

$$V_{L-L_RMS} = \frac{\sqrt{3}}{\sqrt{2}} V_{p_new} = \frac{\sqrt{3}}{\sqrt{2}} \times 540 \approx 661.36 \text{ V}$$

$$n_{new} = 1500 \times \frac{661.36}{180} \approx 5511.33 \text{ rpm}$$

iv) Based on iii), add a third-order harmonic component with an amplitude equal to ⅙ of the fundamental modulating signal, as illustrated in Figure 1. Recalculate the modulation index and explain the benefit of third-harmonic injection. [3]

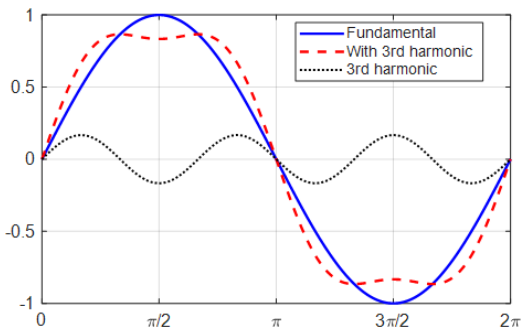


Figure 1. Fundamental modulating signal and its third harmonic-injected waveform

After adding the harmonic component, the function becomes

$$m(\omega t) = m(\theta) = M\left(\sin\theta + \frac{1}{6}\sin 3\theta\right)$$

To find the peak, take derivative with respect to θ

$$\frac{dm}{d\theta} = M\left(\cos\theta + \frac{1}{2}\cos 3\theta\right) = 0$$

from which yields

$$\theta = 60°, 120°$$

The value of m at the peaks is

$$m(60°) = m(120°) = M\left(\sin\theta + \frac{1}{6}\sin 3\theta\right) = \frac{\sqrt{3}}{2}M = 0.866M < M$$

[2]

It is clear that after third-order harmonic injection, the peak value of the modulator is smaller than the peak value without third-order harmonic injection, which means that the third-order harmonic injection allows for higher margins of modulation. [1]

Question 3 – Power Flow in Lines and Cables. This question is worth 25 marks.

a) Consider the characteristics of single-phase overhead lines.

i) An overhead line operates at 50 Hz. Assume the line is lossless and is represented only by its series inductive reactance (resistance and shunt capacitance neglected). The magnitudes of the sending-end and receiving-end voltages are both maintained at 400 kV. The line inductance is 100 μH/km. If the maximum transmittable real power is 20 GW, determine the line length. [3]

Use standard equation

$$P = \frac{V_S V_R}{X}\sin\delta$$

The maximum power is achieved when the load angle equals 90°, therefore

$$P_{\max} = \frac{V_S V_R}{X}$$

Hence

$$X = \frac{V_S V_R}{P_{\max}} = \frac{(400 \times 10^3)^2}{20 \times 10^9} = 8\Omega$$

Also,

$$X = 2\pi f L$$

so the total inductance is

$$L = \frac{X}{2\pi f} = \frac{8}{2\pi \times 50} = 0.02546 \text{ H}$$

The line inductance per km is $L' = 100$ μH/km

Therefore the line length is

$$l = \frac{L}{L'} = \frac{0.02546}{100 \times 10^{-6}} = 254.65 \text{ km}$$

ii) For the line in i), under the same assumptions, if the allowable maximum current is 16.2 kA, determine the load angle δ corresponding to this current limit. [3]

Let the magnitude of the voltage phasor difference between the sending and receiving ends be ΔV.
According to the cosine rule for phasors:

$$\Delta V^2 = V_S^2 + V_R^2 - 2V_S V_R \cos(\delta)$$

Since $V_S = V_R = 400$ kV:

$$\Delta V^2 = 2(400 \times 10^3)^2(1 - \cos(\delta))$$

The voltage difference is caused by the line reactance, thus $\Delta V = I_{\max} X_{SR}$.
Given the maximum allowable current $I_{\max} = 16.2$ kAand $X_{SR} = 8\ \Omega$:

$$\Delta V = 16.2 \times 10^3 \times 8 = 129.6 \text{ kV}$$

Substituting ΔVinto the cosine rule equation:

$$(129.6 \times 10^3)^2 = 2(400 \times 10^3)^2(1 - \cos(\delta))$$
$$1 - \cos(\delta) = \frac{129.6^2}{2 \times 400^2} \approx 0.052488$$
$$\cos(\delta) \approx 0.9475$$
$$\delta = \arccos(0.9475) \approx 18.64°$$

iii) Series compensation is an effective way to increase the loadability of a transmission line. By inserting a capacitor in series with an inductive transmission line, the total reactance can be reduced. Based on the line parameter in i), calculate the value of the series capacitor that reduces the total reactance of the transmission system (line + capacitor) to half of the original line reactance. [3]

To reduce the total reactance of the line to half, the new total reactance should be

$$X_{\text{total}} = \frac{8}{2} = 4\ \Omega$$

The total reactance after adding the series capacitor is

$$X_{\text{total}} = X_{SR} - X_C$$
$$X_C = 8 - 4 = 4\ \Omega$$

The capacitive reactance formula is

$$X_C = \frac{1}{2\pi f C}$$
$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi \times 50 \times 4} = \frac{1}{400\pi} \approx 795.77\ \mu\text{F}$$

iv) Based on iii), calculate the maximum transmittable real power if the series capacitor is added, and explain the impact of series compensation in increasing the loadability of the line. [4]

The maximum transmittable real power is inversely proportional to the total series reactance.

$$P_{\max,\text{new}} = \frac{V_S V_R}{X_{\text{total}}} \sin(90°) = \frac{(400 \times 10^3)^2}{4} = 40 \text{ GW}$$

Impact explanation: By inserting a capacitor in series with the line, the equivalent inductive reactance of the transmission system is reduced. This directly increases the maximum active power transfer limit, effectively increasing the loadability of the line.

v) An overhead line is supplied at 15 kV. The voltage at the receiving end is 14.3 kV. The real power flowing into the line at the sending end is 11.0 MW, and the sending-end power factor is 0.90 leading. The sending-end voltage phasor leads the receiving-end voltage by 20°. Estimate the series impedance of the line. [4]

Calculate the sending-end apparent power using the active power $P_S = 11.0$ MWand leading power factor $pf = 0.90$:

$$S_S = \frac{P_S}{pf} = \frac{11.0}{0.90} \approx 12.22 \text{ MVA}$$

Since the power factor is leading, the reactive power $Q_S$is negative (capacitive):

$$Q_S = -S_S \sin(\arccos(0.90)) = -12.22 \times 0.4359 \approx -5.327 \text{ MVar}$$

Taking $V_S$as the reference phasor (angle 0°), calculate the sending-end current phasor:

$$I_S = \left(\frac{S_S}{V_S}\right)^* = \left(\frac{11.0 \text{ MW} - j5.327 \text{ MVar}}{15 \text{ kV}}\right)^* = 733.3 + j355.1 \text{ A}$$

The receiving-end voltage phasor $V_R$lags $V_S$by 20°:

$$V_R = 14.3 \text{ kV}\angle -20° = 14300(\cos(-20°) + j\sin(-20°))$$
$$= 13437.6 - j4890.8 \text{ V}$$

The voltage drop across the line is:

$$\Delta V_{\text{phasor}} = V_S - V_R = 15000 - (13437.6 - j4890.8)$$
$$= 1562.4 + j4890.8 \text{ V}$$

Calculate the series impedance:

$$Z = \frac{\Delta V_{\text{phasor}}}{I_S}$$

$$Z = \frac{1562.4 + j4890.8}{733.3 + j355.1} = \frac{5134.4\angle 72.28°}{814.8\angle 25.84°} = 6.301\angle 46.44°$$
$$= 4.34 + j4.57 \ \Omega$$

b) Consider the characteristics of single-phase overhead lines suppling an impedance load, as shown in the figure below, and answer these questions.

i) The overhead line has an impedance of 0.5 + j4.0 Ω. The voltage at the sending end is 1300 V. A pure resistive load R_L is connected at the receiving end. If the magnitude of the receiving-end terminal voltage is 1218V, determine the value of R_L. [3]

The line impedance is

$$Z_{\text{line}} = 0.5 + j4.0\ \Omega$$

The total impedance after connecting the pure resistive load is

$$Z_{\text{total}} = (0.5 + R_L) + j4.0\ \Omega$$

The magnitude of the current flowing through the line and load is:

$$|\ I\ | = \frac{|\ V_S\ |}{|\ Z_{\text{total}}\ |} = \frac{1300}{\sqrt{(0.5+R_L)^2+4.0^2}}$$

The receiving-end terminal voltage is given as

$$|\ V_R\ | = |\ I\ |\ R_L = 1218\ \text{V}$$

thus

$$1218 = \frac{1300R_L}{\sqrt{(0.5+R_L)^2+16}}$$

Squaring both sides and letting

$$k = \left(\frac{1300}{1218}\right)^2 \approx 1.139$$

gives

$$1.139R_L^2 = (0.5 + R_L)^2 + 16 = R_L^2 + R_L + 16.25$$
$$0.139R_L^2 - R_L - 16.25 = 0$$

Solving this quadratic equation and taking the positive root:

$$R_L = \frac{1 + \sqrt{1 - 4(0.139)(-16.25)}}{2(0.139)} \approx 15.0\ \Omega$$

ii) For the line in i), consider a pure resistive load R_{L} connected at the receiving end. Calculate the maximum power that can be supplied to the load. (Hint: find the power delivered to R_{L} as a function of R_{L} and identify the maximum value of the function.) [5]

Assuming the load resistance is R_L, the current flowing through the transmission line and the load is expressed as:

$$|\ I\ | = \frac{V_S}{|\ \bar{Z}_S + \bar{Z}_L\ |} = \frac{V_S}{\sqrt{(R_S+R_L)^2+X_S^2}}$$

The active power consumed by the load is expressed as:

$$P_L = I^2R_L = \frac{V_S^2R_L}{(R_S+R_L)^2+X_S^2} = \frac{V_S^2}{R_L + \frac{R_S^2 + X_S^2}{R_L} + 2R_S}$$

Therefore, the load power reaches its maximum when

$$R_L = \sqrt{R_S^2 + X_S^2}$$
$$R_L = \sqrt{0.5^2 + 4.0^2} = \sqrt{16.25} \approx 4.03\ \Omega$$

The corresponding maximum active power is calculated as:

$$P_{L\max} = \frac{V_S^2}{2\sqrt{R_S^2 + X_S^2} + 2R_S} = \frac{1300^2}{2 \times 4.03 + 2 \times 0.5} = \frac{1690000}{9.06}$$
$$\approx 186.53 \text{ kW}$$

Question 4 – Switch-Mode Power Supplies. This question is worth 30 marks.

a) A Buck switch-mode power supply (SMPS) is to be designed for an input voltage of 36 V and an output voltage of 12 V. The converter must deliver a maximum output power of 48 W, and the peak-to-peak output voltage ripple must not exceed 60 mV.

Ignore the MOSFET conduction losses. The turn-on and turn-off energy losses of the MOSFET are 2.5 μJ and 3.5 μJ respectively when operated at 1 A, and scale linearly with current. The converter initially operates at a switching frequency of 40 kHz.

i) Design the inductance such that, at maximum output power, the peak-to-peak inductor current ripple ΔI_L does not exceed 20% of the average output current. [3]

Given

$$V_{in} = 36 \text{ V}, V_o = 12 \text{ V}$$
$$P_o = 48 \text{ W}, f_s = 40 \text{ kHz}$$
$$\Delta V_o \leq 60 \text{ mV}$$

Duty ratio

$$D = \frac{V_o}{V_{in}} = \frac{12}{36} = \frac{1}{3}$$

Output current

$$I_o = \frac{P_o}{V_o} = \frac{48}{12} = 4 \text{ A}$$

Ripple requirement

$$\Delta I_L \leq 0.2 I_o$$
$$\Delta I_L \leq 0.8 \text{ A}$$

Buck ripple equation

$$\Delta I_L = \frac{(V_{in} - V_o)D}{L f_s}$$

Substitute values

$$L = \frac{(36 - 12) \times \frac{1}{3}}{0.8 \times 40 \times 10^3}$$

$$L_{min} = 250 \mu H$$

ii) Assume a film capacitor is used at the output, with sufficiently small ESR such that the ESR-induced ripple can be neglected. Design the minimum output capacitance such that the peak-to-peak output voltage ripple ΔV_o does not exceed 60 mV. [3]

Capacitive ripple

$$\Delta V_o = \frac{\Delta I_L}{8 f_s C}$$
$$C = \frac{\Delta I_L}{8 f_s \Delta V_o}$$

Substitute values

$$C = \frac{0.8}{8 \times 40 \times 10^3 \times 0.06}$$

$$C_{min} = 41.7\mu F$$

iii) If an electrolytic output capacitor is used, neglect capacitive ripple and determine the maximum ESR allowed such that for the peak-to-peak output voltage ripple ΔV_o does not exceed 60 mV. [3]

Ripple due to ESR

$$\Delta V_o = \Delta I_L \cdot ESR$$
$$ESR = \frac{\Delta V_o}{\Delta I_L}$$
$$ESRC_{max} = 75m\Omega$$

iv) At maximum output power, verify whether the initial switching frequency of 40 kHz satisfies the design requirement that the MOSFET switching loss must not exceed 1.0 W. [3]

Inductor current limits

$$I_{min} = 4 - 0.4 = 3.6 \text{ A}$$
$$I_{max} = 4 + 0.4 = 4.4 \text{ A}$$

Switching energies scale linearly with current

$$E_{on} = 2.5 \times 3.6 = 9\mu J$$
$$E_{off} = 3.5 \times 4.4 = 15.4\mu J$$

Energy per cycle

$$E_{sw} = 24.4\mu J$$

Switching loss

$$P_{sw} = E_{sw} f_s = 0.98W < 1W$$

Therefore, requirement is satisfied.

v) Then consider reducing the switching frequency to 20 kHz. Recalculate:

- the MOSFET switching loss,
- the inductance required to keep the same ΔI_L as in i),
- the capacitance required to keep the same ΔV_o as in ii).

Comment on the design implications. [4]

Switching loss

$$P_{sw} = E_{sw} f_s$$
$$P_{sw} = 24.4\mu J \times 20\text{kHz} = 0.49\text{W}$$

Inductor value

$$L = \frac{(V_{in} - V_o)D}{\Delta I_L f_s}$$
$$L = \frac{24 \times \frac{1}{3}}{0.8 \times 20 \times 10^3} = 500\mu H$$

Capacitor value

$$C = \frac{\Delta I_L}{8 f_s \Delta V_o}$$

$$C = \frac{0.8}{8 \times 20 \times 10^3 \times 0.06} = 83.3\mu F$$

Comment
Reducing switching frequency:
reduces switching loss
requires larger inductance
requires larger capacitance
Therefore, converter size and cost increase, reducing power density.

vi) For a purely resistive load, derive the transfer function from the switch-node voltage v_{sw} to the output voltage v_o of the output RLC network and explain, in the frequency domain, how it behaves as a low-pass filter that attenuates switching ripple. [4]

Parallel load impedance

$$Z_p = R \parallel \frac{1}{sC} = \frac{R}{1 + sRC}$$

Transfer function

$$G(s) = \frac{v_o(s)}{v_{sw}(s)} = \frac{Z_p}{sL + Z_p}$$

Substitute

$$G(s) = \frac{R}{LRCs^2 + Ls + R}$$

This represents a second-order low-pass filter.
Low frequencies pass with little attenuation.
High-frequency switching components are attenuated.
Roll-off ≈ −40 dB/decade.
Thus the LC network suppresses switching ripple.

b) A second SMPS of Buck-Boost form is required to produce a -20 V output from the 120 V input. The converter operates at a switching frequency of 50 kHz in discontinuous conduction mode (DCM).

At an output power of 200 W, the duty cycle is 0.2.

The MOSFET to be used has the following switching characteristics:

Turn-on energy loss E_{on} = 12.0 μJ

Turn-off energy loss E_{off} = 18.0 μJ

The MOSFET junction temperature must not exceed 70°C. The ambient temperature is 40°C, and the effective junction-to-air thermal resistance is

$$R_{\text{th}} = 15 \text{ K/W}$$

i) Calculate the energy that must be stored in the inductor at the end of the on-time in each switching cycle in order to transfer 200 W output power. [2]

Given

$$V_{in} = 120\ \text{V}$$
$$V_o = -20\ \text{V}$$
$$P_o = 200\ \text{W}$$
$$f_s = 50\text{kHz}$$
$$D = 0.2$$

In DCM all stored energy is delivered each cycle

$$W_L = \frac{P_o}{f_s}$$
$$W_L = \frac{200}{50 \times 10^3} = 4mJ$$

ii) Calculate the required inductance, and the inductor current at the end of the on-time. [4]

Switching period

$$T_s = 20\mu s$$

On-time

$$t_{on} = DT_s = 4\mu s$$

Energy stored

$$W_L = \frac{1}{2}LI_{pk}^2$$

Peak current

$$I_{pk} = \frac{V_{in}t_{on}}{L}$$

Substitute

$$L = \frac{V_{in}^2 t_{on}^2}{2W_L} = 28.8\mu H$$

Peak current

$$I_{pk} = 16.7A$$

iii) Using the operating condition above, determine the maximum allowable MOSFET on-state resistance $R_{\text{DS(on)}}$ so that the thermal design requirement is satisfied. [4]

Thermal limit

$$P_{max} = \frac{T_j - T_a}{R_{th}}$$
$$P_{max} = \frac{70 - 40}{15} = 2W$$

Switching loss

$$P_{sw} = (E_{on} + E_{off})f_s$$
$$P_{sw} = 30\mu J \times 50kHz = 1.5W$$

Remaining conduction loss

$$P_{cond} = 0.5 \text{ W}$$

MOSFET RMS current

$$I_{MOS,rms}^2 = \frac{DI_{pk}^2}{3}$$
$$I_{MOS,rms}^2 = 18.5$$

Conduction loss

$$P_{cond} = I_{MOS,rms}^2 R_{DS(on)}$$
$$R_{DS(on)} = \frac{0.5}{18.5} = 27m\Omega$$